

DevOps Series

Hardening of Parabola

Every computer offers security measures to isolate it from outside attacks. In this 21st article in the DevOps series, we will learn how to harden, automate and verify a Parabola GNU/Linux-libre system using Ansible.



Parabola GNU/Linux-libre is a free software GNU/Linux distribution based on packages from Arch Linux, but without the binary blobs. It respects the freedom of users and is endorsed by the Free Software Foundation. It is extremely lightweight and simple in design. The distribution follows a rolling release, but also provides installation media in the ISO format. The Pacman package manager is used, and the software packages are available for i686, x86_64 and armv7h architectures. The Linux-libre kernel is used instead of the generic Linux kernel. The official IRC channel is *#parabola* on *irc.freenode.net*, and the website of the project is <https://www.parabola.nu/>.

Setup

A Parabola GNU/Linux-libre (x86_64) 2018.06.02 ISO is used to set up a guest virtual machine (VM) with KVM/QEMU. The host system is also a Parabola GNU/Linux-libre x86_64 system, and Ansible is installed using the distribution package manager. The version of Ansible used is 2.6.0, as is indicated below:

```
$ ansible --version
ansible 2.6.0
  config file = /etc/ansible/ansible.cfg
```

```
configured module search path = ['/home/guest/.ansible/
plugins/modules', '/usr/share/ansible/plugins/modules']
ansible python module location = /usr/lib/python3.6/site-
packages/ansible
```

```
executable location = /usr/bin/ansible
python version = 3.6.5 (default, May 11 2018, 04:00:52)
[GCC 8.1.0]
```

The Ansible inventory, playbook and configuration files are created on the host system as follows:

```
ansible/inventory/kvm/
  /playbooks/configuration/
  /files/
```

The *inventory/kvm/inventory* file contains the following:

```
parabola ansible_host=192.168.122.128 ansible_connection=ssh
ansible_user=parabola ansible_password=password
```

The *sudo* package needs to be installed in the guest Parabola VM. A *parabola* user is created in the guest VM, and